

Solar O appendix

Sun · O · downloadable appendix

O.json v1.9 · 7 visible products · 0 quarantined entries

products

Independent admitted products; no combined headline abundance.

Highlighted Products

No adopted highlighted product has been selected. See the independent products below.

Error-bar Forest Plot

red-optical — iag_fts_solar_atlas solar_iag A(O) dex; axis 8.77-9.04		
SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=6		8.899 ±0.044 stat ±0.043 syst
indicator not supplied · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=2		8.901 ±0.104 stat ±0.043 syst
red-optical — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(O) dex; axis 8.77-9.04		
SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=6		8.894 ±0.025 stat ±0.043 syst
indicator not supplied · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=2		8.912 ±0.070 stat ±0.043 syst
red-optical — kpno_solar_atlas solar_kpno_molecfits_corrected A(O) dex; axis 8.77-9.04		
SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=6		8.873 ±0.033 stat ±0.043 syst
indicator not supplied · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=2		8.892 ±0.085 stat ±0.043 syst
red-optical axis: 8.77 · 8.83 · 8.90 · 8.97 · 9.04 A(O) dex		
VIS — harps_solar_harps_molecfits_corrected A(O) dex; axis 8.36-9.26		
FORB-OI_6300 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=1		8.810 ±0.421 stat ±0.006 syst

VIS axis: 8.36 · 8.58 · 8.81 · 9.04 · 9.26 A(O) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence.

Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

Solar oxygen context

Oxygen is traced by forbidden [O I], permitted O I and molecular OH indicators where available. Blends, departures from LTE and atmospheric structure affect these families differently, so their results retain separate identities.

[C/O indicator strategy](#) · [Methodology](#)

Product provenance and caveats

SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "line-to-line scatter", "gf_pool_graded": true, "gf_rung_summary": "gf rung 2 (gf scale (cited NIST class))", "grade": "Reference Grade", "n_excluded": 0, "n_lines": 6, "provenance": {"artifact_mtime": "2026-09-12T20:22:50Z", "copied_to": "data/results/band_products/OI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_SET-AGSS21_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:50Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/OI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_SET-AGSS21_products.csv", "sha256": "215a07c0904c1913044f2082d9f5611a968e85e7a733ee5c4d53b9aa80344316"}, "sigma_stat": 0.044, "sigma_syst": 0.0429, "stat_basis": "measured — RMS of the random terms, 0.04403 dex at n_lines=6"}

indicator not supplied · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "line-to-line scatter", "grade": "Codex Grade", "n_excluded": 0, "n_lines": 2, "provenance": {"artifact_mtime": "2026-09-12T20:07:34Z", "copied_to": "data/results/band_products/OI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:50Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/OI_6910_9199_iag_fts_solar_atlas_solar_iag_SYNTH_products.csv", "sha256": "d7166bab3667b9190b50f4eda630872e3e2f5fa08c8f97dc089b598a3de6bc96"}, "sigma_stat": 0.1045, "sigma_syst": 0.0429, "stat_basis": "measured — RMS of the random terms, 0.10450 dex at n_lines=2"}

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SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "line-to-line scatter", "gf_pool_graded": true, "gf_rung_summary": "gf rung 2 (gf scale (cited NIST class))", "grade": "Reference Grade", "n_excluded": 0, "n_lines": 6, "provenance": {"artifact_mtime": "2026-09-12T20:21:50Z", "copied_to": "data/results/band_products/OI_6910_9199_kpno_solar_atlas_solar_kpno_molecfit_corrected_SYNTH_SET-AGSS21_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:52Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/OI_6910_9199_kpno_solar_atlas_solar_kpno_molecfit_corrected_SYNTH_SET-AGSS21_products.csv", "sha256": "2a4cc12be7fffb00e00b53423adfc15fe7d8b9e47c89fec753290602fa637e95"}, "sigma_stat": 0.0332, "sigma_syst": 0.0429, "stat_basis": "measured — RMS of the random terms, 0.03325 dex at n_lines=6"}

indicator not supplied · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": "line-to-line scatter", "grade": "Codex Grade", "n_excluded": 0, "n_lines": 2, "provenance": {"artifact_mtime": "2026-09-12T20:05:10Z", "copied_to": "data/results/band_products/OI_6910_9199_kpno_solar_atlas_solar_kpno_molecfits_corrected_SYNTH_products.csv", "host": "Sirius", "ingested_at": "2026-09-12T23:16:52Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/OI_6910_9199_kpno_solar_atlas_solar_kpno_molecfits_corrected_SYNTH_products.csv", "sha256": "bf1081efe806cb25c8c11e2bd87d6d1e84539b7b99634f61e64182df7d229737"}, "sigma_stat": 0.0845, "sigma_syst": 0.0429, "stat_basis": "measured — RMS of the random terms, 0.08450 dex at n_lines=2"}

FORB-OI_6300 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied: {"display": "Synth · 1D-LTE", "dominant_term": null, "grade": "Codex Grade", "n_excluded": 0, "n_lines": 1, "provenance": {"artifact_mtime": "2026-09-13T04:18:40Z", "copied_to": "data/results/band_products/OI_FORB_OI_6300_harps_solar_harps_molecfits_corrected_SYNTH_FORB-OI_6300_products.csv", "host": "Sirius", "ingested_at": "2026-09-13T04:21:45Z", "path": "/home/damienabraxas/scratch/rya1214/data/results/band_products/OI_FORB_OI_6300_harps_solar_harps_molecfits_corrected_SYNTH_FORB-OI_6300_products.csv", "sha256": "8f3b5cce06db71745af92f108693faeb0999190b208abfb2b886444cf290ea45"}, "sigma_stat": 0.421, "sigma_syst": 0.006, "stat_basis": "measured — 1 sigma from the chi2 curvature of THIS band's fit (pipeline.fit_constraint.curvature_sigma, rescaled to red_chi2 = 1); NaN where the curvature was not measurable, never 0 and never the retired 1.000 clip"}

Open provenance decisions

O I 615.8nm: □ DISAGREES — reported to Ryan, not resolved. . Store log gf -0.409 (); summed pull -0.2966 (B+).

O I 630.0nm: □ DISAGREES — reported to Ryan, not resolved. □ store claims A, pull publishes B+. Store log gf -9.717 (A); summed pull -9.7744 (B+).

OPEN. Tachiev and NIST differ systematically by -0.016 dex on the 777 triplet and -0.005 dex on 8446. AGSS21 names Tachiev for N i only and states no O i gf source. Both values are reported in oi_tachiev_vs_nist.csv; NIST is what the store already carried and this script does not change it. Ryan decides.

[Adjudication record](#) · [Indicator gf checks](#)

Product visibility audit

Every feed entry is accounted for. Quarantined and archived entries retain their source disposition; their numeric fit outputs are available only in the labelled source download.

Withheld entries and source reasons (0)

Download the evidence

[Download PDF — Solar O appendix](#)

[Source feed — includes quarantined fit outputs, not adopted abundances](#)

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

Reproducibility

Generated from the pinned CNO campaign snapshot. Source commit 22ea65678043e016b12fb0defa596c10a200df88

Feed O.json v1.9 · updated 2026-09-25T19:39:46Z

Generated 2026-09-25T21:53:06-06:00 (source commit timestamp)

Feed SHA-256 fbe537833fcc795b6e7cfdafa2fe70c7c3c9298a43faed59df8e76c30aa14ff7

Generator scripts/generate_cno_publication.py v1.0.0

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar O atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [1], [2].

Atomic data and gf authorities. The recorded O I/O II base pedigree is Wiese-Fuhr-Deters / Opacity Project theory. Per-transition exceptions and the open O I source decisions remain explicit. Sources: [3], [4], [5].

Atomic data and gf authorities. O I triplet source record: NIST (this script) — Tachiev NOT adopted; OPEN — referred to Ryan (RYA-1214 EOS) Sources: [3], [6].

Model and grid sources. Registered O I line/grid lineage: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — line-by-line 3D-NLTE grids built on the Amarsi+2018 O I model atom. The current feed treatment remains authoritative; a grid reference does not imply a correction was applied. Sources: [7], [8].

Atomic data and gf authorities. Atomic O I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4], [5].

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [9].

Special diagnostics. The synthesis sidecar names Johansson's Ni I blend data, Storey & Zeippen's forbidden-line gf lineage and Caffau's CO5BOLD comparison anchor. The feed entry remains 1D-LTE and quarantined; the sidecar's cited anchor is not promoted to a measured abundance. Sources: [10], [11], [12].

Product-to-source index

O-products-361df42bd1f9 · products · red-optical · solar_iag · SET-AGSS21 · 1D-LTE · not supplied · [1], [3], [4], [5], [9]

O-products-945c945b31da · products · red-optical · solar_iag · not supplied · 1D-LTE · not supplied · [1], [3], [4], [5], [9]

O-products-e2dedb7ba833 · products · red-optical · solar_kpno_kurucz2005_corrected · SET-AGSS21 · 1D-LTE · not supplied · [1], [3], [4], [5], [9]

O-products-eadb1d33ffa2 · products · red-optical · solar_kpno_kurucz2005_corrected · not supplied · 1D-LTE · not supplied · [1], [3], [4], [5], [9]

O-products-da700d54e88d · products · red-optical · solar_kpno_molecfit_corrected · SET-AGSS21 · 1D-LTE · not supplied · [1], [3], [4], [5], [9]

O-products-255dd9842275 · products · red-optical · solar_kpno_molecfit_corrected · not supplied · 1D-LTE · not supplied · [1], [3], [4], [5], [9]

O-products-a69ed2a177ae · products · VIS · solar_harps_molecfit_corrected · FORB-OI_6300 · 1D-LTE · not supplied · [1], [3], [4], [5], [9], [10], [11], [12]

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. [source record](#)

[2] — Amarsi et al. 2021, A&A 656 A113 DOI: 10.1051/0004-6361/202141384. [source record](#)

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. [source record](#)

[4] — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) [source record](#)

[5] — Wiese, W. L.; Fuhr, J. R.; Deters, T. M. (1996). Atomic Transition Probabilities of Carbon, Nitrogen, and Oxygen: A Critical Data Compilation. JPCRD Monograph 7. [source record](#)

[6] — Tachiev & Froese Fischer 2002, A&A 385, 716 DOI: 10.1051/0004-6361/20011816. [source record](#)

Model and grid sources

[7] — Amarsi, A. M.; Nissen, P. E.; Skuladottir, A. (2019). Carbon, oxygen, and iron abundances in disk and halo stars. A&A 630, A104 DOI: 10.1051/0004-6361/201936265. [source record](#)

[8] — Amarsi, Barklem, Asplund, Collet & Zatsarinny 2018, A&A 616, A89 DOI: 10.1051/0004-6361/201832770. [source record](#)

[9] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. [source record](#)

Special diagnostics

[10] — Johansson, S.; Litzen, U.; Lundberg, H.; Zhang, Z. (2003). Experimental f-Value and Isotopic Structure for the Ni I Line Blended with [O I] at 6300 Å. ApJ 584, L107 DOI: 10.1086/374037. [source record](#)

[11] — Storey, P. J.; Zeippen, C. J. (2000). Theoretical values for the [O III] 5007/4959 line-intensity ratio and homologous cases. MNRAS 312, 813 DOI: 10.1046/j.1365-8711.2000.03184.x. [source record](#)

[12] — Caffau, E.; Ludwig, H.-G.; Steffen, M.; Livingston, W.; Bonifacio, P.; Malherbe, J.-M.; Doerr, H.-P.; Schmidt, W. (2015). The photospheric solar oxygen project. III. Investigation of the centre-to-limb variation of the 630 nm [O I]-Ni I blend. A&A 579, A88. [source record](#)